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application needs. The unused gates continue to consume power, resulting in significant power inefficiency, too. While adaptive power management techniques help to reduce power consumed by unused gates, the effectiveness of such techniques is limited due to the coarse granularity

at which these must be applied, as well as significant implementation overheads such as domain isolation and state retention. These techniques also worsen area inefficiency.

Approaches

One approach to significantly increase the area and power efficiency of a microprocessor for a given application is to eliminate all logic in the microprocessor IP core that will not be used by the application. Eliminating logic that is not used by an application can produce a design tailored to the application—a bespoke processor—that has significantly lower area and power requirements than the original microprocessor IP that targets an arbitrary application.

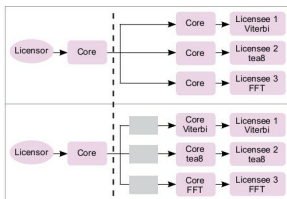


Fig. 1: General-purpose processors are overdesigned for a specific application (top). The bespoke processor design methodology allows a microprocessor IP licensor or licensee to target different applications efficiently without additional software or hardware development cost (bottom)

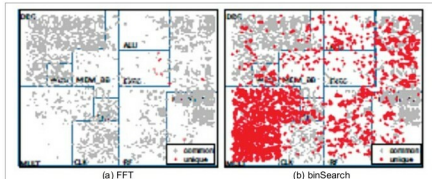


Fig. 2: Gates not toggled by two applications—FFT and binSearch—for profiling inputs. Gray gates are not toggled by either application. Red gates are unique untoggled gates for each application

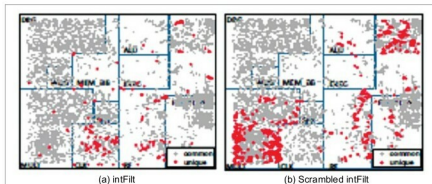


Fig. 3: Gates not toggled by (a) intFill and (b) scrambled intFill for the same input set. Red gates are uniquely not toggled by that application. Gray gates are not toggled by either application. Red gates are unique untoggled gates for each application. Even though the applications use the same set of instructions and control flow, the gates that they exercise are different